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# CONSIDERATIONS FOR SHIPPING AND STORAGE OF 1-PHASE COLD PLATE LIQUID COOLED RACKS AND SERVERS.

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## Version History

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## Executive Summary

Liquid cooled AI infrastructure requires individual racks and servers to be shipped and stored correctly to the final deployment site. One important aspect is whether the liquid cooled racks and servers should be shipped filled, partially filled, dry or semi dry. Each scenario has its benefits and drawbacks which are evaluated in detail in this document. We also consider best practices to reduce unwanted failures during shipping and storage.

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## 1. Introduction

This whitepaper provides guidelines for the shipment and storage of liquid-cooled IT servers and racks. It addresses both technical risks and operational considerations, while offering best practices to mitigate potential issues such as thermal expansion, corrosion, and coolant degradation. Upon review of this document, data center deployment teams will have a better understanding of the best practices for the transportation and storage of liquid-cooled equipment.

## 2. Compliance with Open Compute Project Tenets

### 2.1. Openness

This whitepaper lowers the burden of entry for deploying liquid cooled hardware into both new and retrofitted datacenters. It can be used by a wide variety of datacenter operators, MEP (Mechanical, Electrical and Plumbing) consultants, engineering staff and others. The procedures are designed to optimize the shipment of liquid cooled collateral to the data center as well as storing them when needed.

### 2.2. Efficiency

This whitepaper provides practical guidance to ensure liquid cooled servers and racks are shipped to the datacenter via the most efficient path for datacenter operational teams. Considerations include approaches to minimizing delays from incorrect shipment methods and damaged hardware. Additional efficiency improvements include utilizing storage environments where available to streamline the deployment model.

### 2.3. Impact

With the transition from air to liquid cooling in datacenters, the shipping and storage of servers and racks to the deployment sites and the associated best practices are important to address. This document will allow better operational planning and minimize downtime from racks and servers that are delivered to the data center site in the incorrect state.

### 2.4. Scale

Liquid-cooled deployments for AI applications are highly scalable, supporting clusters ranging from small data centers (fewer than 100 racks) to large data centers (more than 100 racks) and AI Factories (exceeding 1,000 racks). Across all deployment scales, it is essential that racks and servers are shipped in a manner that ensures they are free from damage and leaks, arriving fully prepared for seamless integration and operation within the data center environment.

### 2.5. Sustainability

The adoption of liquid cooling in datacenters enables the efficient use of a liquid medium as the heat transfer fluid, which facilitates the recovery of waste heat generated by AI hardware. This approach unlocks new opportunities for implementing heat reuse solutions across a wide range of industrial applications. By capturing heat at higher temperatures and with greater efficiency than traditional air-cooled systems, liquid-cooled datacenters can supply recovered thermal energy for district heating, industrial processes, greenhouse operations, or domestic hot water, thereby enhancing overall energy utilization and supporting sustainability objectives.

## 3. Overview of Shipment and Storage Methods.

**Table 1:** Summary of various shipment methods, highlighting their associated technical risks and operational burdens. Note that certain detailed procedures, such as bake drying, nitrogen purging, pressure checks, and leak testing are not included in this table.

| Shipment                    | Wet                         | Partial Wet        | Dry                         | Semi Dry          |
|-----------------------------|-----------------------------|--------------------|-----------------------------|-------------------|
| Liquid Fill Ratio           | <b>Almost 100%</b>          | <b>90%~95%</b>     | <b>&lt; 10%</b>             | <b>&lt; 50%</b>   |
| User Scenario               | General Inbound or Outbound | Early NPI Inbound  | General Inbound or Outbound | Fast Outbound RMA |
| <b>Technical Risk</b>       |                             |                    |                             |                   |
| Risk of Expansion Damage    | Medium                      | Low                | Low                         | Low               |
| Risk of Corrosion           | Low                         | Medium             | Low                         | High              |
| <b>Operational Burden</b>   |                             |                    |                             |                   |
| Prepare - Drain Liquid      | N                           | Y                  | Y                           | Y                 |
| Prepare - Charge N2         | N                           | N                  | Y                           | N                 |
| Shipment - Temp Control     | Y                           | Y                  | N                           | N                 |
| Before Use - Topoff/Prefill | N                           | Y                  | Y                           | Y                 |
| <b>Storage</b>              | <b>Wet</b>                  | <b>Partial Wet</b> | <b>Dry</b>                  | <b>Semi Dry</b>   |
| <b>Technical Risk</b>       |                             |                    |                             |                   |
| Risk of Expansion           | Medium                      | Low                | Low                         | Low               |
| Risk of Corrosion           | Low                         | High               | Medium                      | High              |
| <b>Operational Burden</b>   |                             |                    |                             |                   |
| Warehouse - Temp Control    | Y                           | Y                  | N                           | N                 |
| Warehouse-Shelf Life        | Long                        | Short              | Medium                      | Short             |
| Warehouse - N2 Renew        | N                           | NA                 | Y                           | NA                |
| Before Use - Prefill        | N                           | Y                  | Y                           | Y                 |

### 3.1. Ship Wet

This category encompasses liquid-cooled IT equipment, including servers and sidecars that are shipped fully filled with coolant (100%). It applies to both inbound and outbound logistics where the hardware is transported in a fully operational, “wet” state. Shipping equipment streamlines operational procedures and is often the most efficient method for preparing large numbers of racks for immediate deployment in datacenters. However, this approach increases the risk of expansion-related damage due to temperature fluctuations during transit. To mitigate this risk, temperature control measures are typically implemented.

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### 3.2. Ship Partial Wet

This category includes liquid-cooled IT equipment such as servers and sidecars that are partially filled with coolant (90–95%). It is typically used for early NPI (New Product Introduction) inbound shipments when the equipment’s structural resilience to expansion-related damage is not yet fully validated. A small percentage of unfilled volume acts as an expansion buffer, reducing the risk of damage due to temperature-induced coolant expansion during transit. Temperature control remains necessary. Upon arrival, additional coolant top-off is required for operational readiness; however, this step imposes a significantly lower operational burden compared to full-scale liquid prefill.

### 3.3. Ship Dry

This category covers liquid-cooled IT equipment including servers and sidecars that are shipped dry, containing less than 10% coolant by volume. It is suitable for general inbound and outbound shipments where the majority of coolant is drained prior to transport.

Shipping dry is the most common approach for delivering liquid-cooled products, particularly for material transfers between component suppliers and system integrators. This method eliminates the risk of expansion-related damage but requires additional procedures both before shipment and after arrival. Typical steps include nitrogen purging and release, shelf life monitoring, and coolant pre-fill upon installation. Shipping dry is recommended when products involve varying or unpredictable coolant types, diverse transportation routes, or uncertain shipment and storage conditions.

### 3.4. Ship Semi Dry

This category pertains to liquid-cooled IT equipment, including servers and sidecars that are shipped semi-dry, retaining less than 50% coolant by volume. It is specifically applicable to expedited outbound RMA (Return Material Authorization) shipments, where equipment is pumped and drained but not fully dried, leaving a significant amount of residual coolant.

The semi-dry shipping method is uniquely suited for rapid RMA scenarios, where minimizing turnaround time is critical for maintaining capacity and reducing spare inventory requirements. By shipping equipment semi-dry, organizations can bypass the more time-consuming procedures associated with dry shipments and avoid the need for temperature-controlled transportation, enabling near-immediate dispatch of returned materials.

### 3.5. Storage

Storage is typically an extension of the shipment process, occurring prior to the equipment's arrival at its final destination and conversion to a production-ready state. As such, similar technical risks and operational requirements apply, with the added challenge of prolonged durations that can increase corrosion risks.

1. Store Wet: Requires temperature control. Over time, corrosion inhibitor levels in the coolant may deplete; studies suggest acceptable limits may be reached in 9-12 months, depending on the copper surface-to-fluid volume ratio. It is recommended to maintain storage temperatures below 25°C (average), minimize temperature fluctuations, and store equipment on a flat surface. Shelf life can be extended by replenishing or circulating fresh coolant to restore inhibitor levels.
2. Store Dry: Requires periodic nitrogen replenishment. Nitrogen protection gradually decays, with studies indicating acceptable pressure limits may be breached in 3-12 months, depending on storage temperature, initial pressure, and cooling system design. Shelf life can be extended by recharging nitrogen to restore protection.

Note: 'Store Partial Wet' and 'Store Semi-Dry' conditions result in significantly increased oxygen and moisture exposure and are not recommended for long-term storage. Units shipped in these states should be converted to either 'Wet' or 'Dry' storage conditions before extended storage.



## 4. Technical Risks

### 4.1 Risk Of Expansion Damage

The risk of expansion during shipment can occur due to changes in temperature that cause the coolant to expand or contract. Within a fixed structure, this can lead to increased internal pressure and potential damage to the system. The thermal variations the equipment is exposed to will depend on the final freight shipment method used. There are four primary methods: air, road (truck), rail, and sea. Each method can also have additional temperature controls in place.

IEC 60721-3-2 [1] *“Classification of groups of environmental parameters and their severities - Transportation and Handling”* has documented the following temperature ranges for each freight method. (See Table 2)

We recommend working with the shipping provider to confirm the temperature ranges the equipment will be exposed to. A test/dummy rack with thermal sensors can also be retrofitted with thermal sensors during a freight shipment to verify the thermal variations are within the expected temperature ranges.

**Table 2 - Temperature range of different transportation approaches**

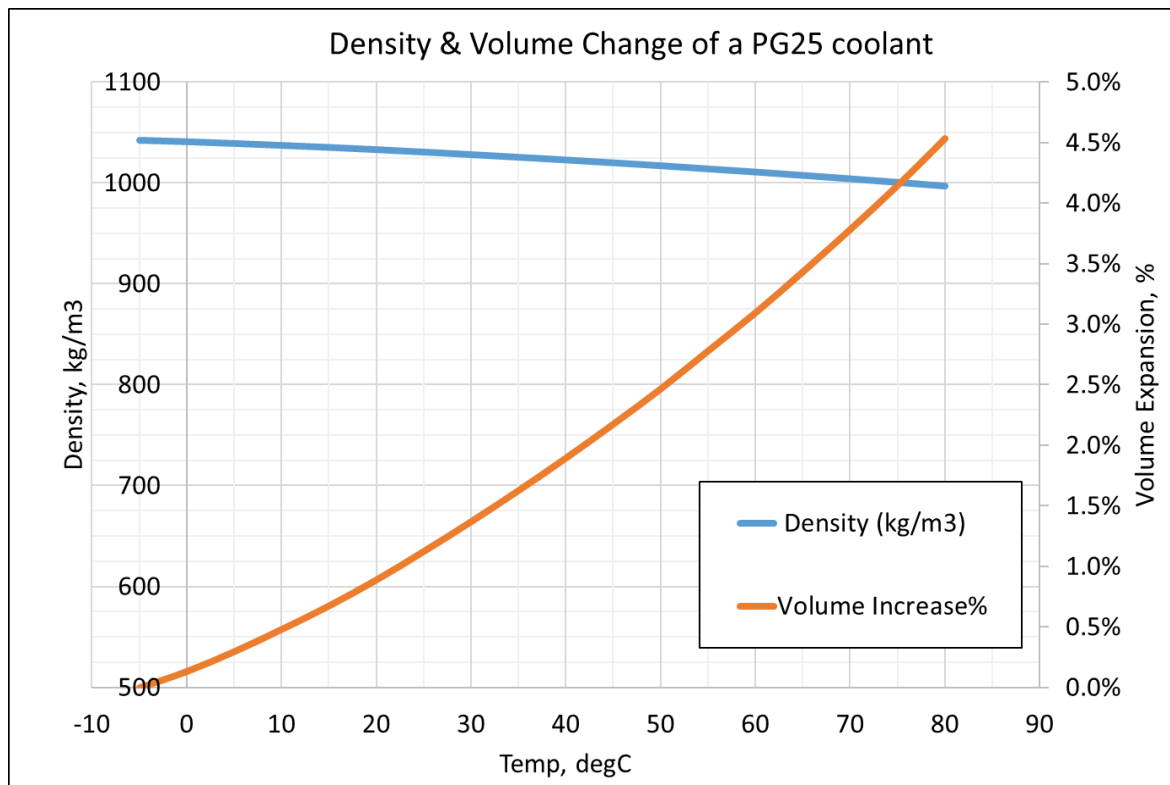
| Shipment Method      | Temperature Range (Standard) | Temperature Range (Controlled) |
|----------------------|------------------------------|--------------------------------|
| Air Freight          | -10°C (14°F) to 40°C (104°F) | 5°C (41°F) to 25°C (77°F)      |
| Road Freight (Truck) | -20°C (-4°F) to 45°C (113°F) | 5°C (41°F) to 25°C (77°F)      |
| Rail Freight         | -20°C (-4°F) to 45°C (113°F) | 5°C (41°F) to 25°C (77°F)      |
| Sea Freight          | -10°C (14°F) to 50°C (122°F) | 5°C (41°F) to 25°C (77°F)      |

\*\* Source - **IEC 60721-3-2** Classification of groups of environmental parameters and their severities - Transportation and Handling

## 4.2. Fluid Expansion

A range of coolant options is available for the deployment of liquid-cooled racks, servers, and sidecars. This document focuses on two primary fluid categories: inhibited propylene glycol PG25 (25 vol.%) and treated water, as these represent the most commonly used coolants in 1-phase liquid-cooled datacenter environments. While other fluids, such as PG55 (55 vol.%) are available for specific shipping and storage scenarios, the scope here is limited to the most prevalent deployment fluids. For technical risks associated with corrosion, refer to Section 4.3.2.

The final selection of shipping fluid should account for the thermal variations the equipment may encounter during transit. Additional considerations are required when shipping outside the recommended temperature range for the chosen fluid. For inhibited propylene glycol, a slush zone exists between the freeze point and burst point, where the fluid begins to crystallize but is not fully frozen. For a 25% propylene glycol solution, this zone typically occurs between -10°C and -15°C. In certain scenarios, end users may utilize the slush zone as a shipping condition to facilitate rapid rack-and-roll deployment of liquid-cooled racks. The authors recommend comprehensive testing of cooling assemblies filled with the final deployment fluid to ensure all components can withstand anticipated thermal variations during shipment.



**Figure 1: PG25 based coolant density reduction and volume increase as temperature increases.**

In addition, it has been observed that PG25 exhibits a volume increase of approximately 2–5% upon freezing, which results in less freeze expansion damage compared to water. However, the risk of localized freezing and subsequent blockage must be considered. When PG25 is exposed to temperatures below its freezing point, the risk of expansion-related damage due to localized freeze and blockage is significantly higher than the volume expansion observed in its liquid state at elevated temperatures.

### 4.3 Fluid/Coolant-Related Concerns During Shipment of Liquid-Cooled Racks

When shipping liquid-cooled racks, coolant stability and potential degradation over time can impact the integrity and performance of the cooling system. The choice between shipping racks

filled with coolant or with nitrogen, as well as the management of any residual fluid, is critical in mitigating these concerns.

#### **4.3.1 Fluid Thermooxidative Breakdown**

For PG25 (25 vol.% propylene glycol), thermooxidative breakdown may occur over time, leading to the formation of acidic by-products. While this process is typically associated with elevated temperatures (above 150°C), gradual pH reduction can still occur at lower temperatures, potentially influencing corrosion mechanisms affecting metals within the cooling loop.

#### **4.3.2 Corrosion**

For wet shipments, the primary concern is whether the reduced pH resulting from thermooxidative breakdown of propylene glycol increases the risk of corrosion. Temperature is a key factor; most shipments occur within a short time frame (<1 month), during which fluid temperatures may reach up to 60°C depending on shipping conditions. For storage periods exceeding one month, it is recommended to maintain temperatures between 20°C and 25°C to minimize corrosion risk.

For dry shipments, the main concern is oxygen exposure to internal metallic surfaces. To mitigate this, manufacturers should ensure proper passivation by circulating a Benzotriazole (BTA)-based solution through the assembly prior to shipment. Subsequently, purging with nitrogen is recommended to minimize residual oxygen within the assembly during transit. Refer to Section 5.3 for additional details on nitrogen filling.

#### **4.3.3 Depletion of Corrosion Inhibitors**

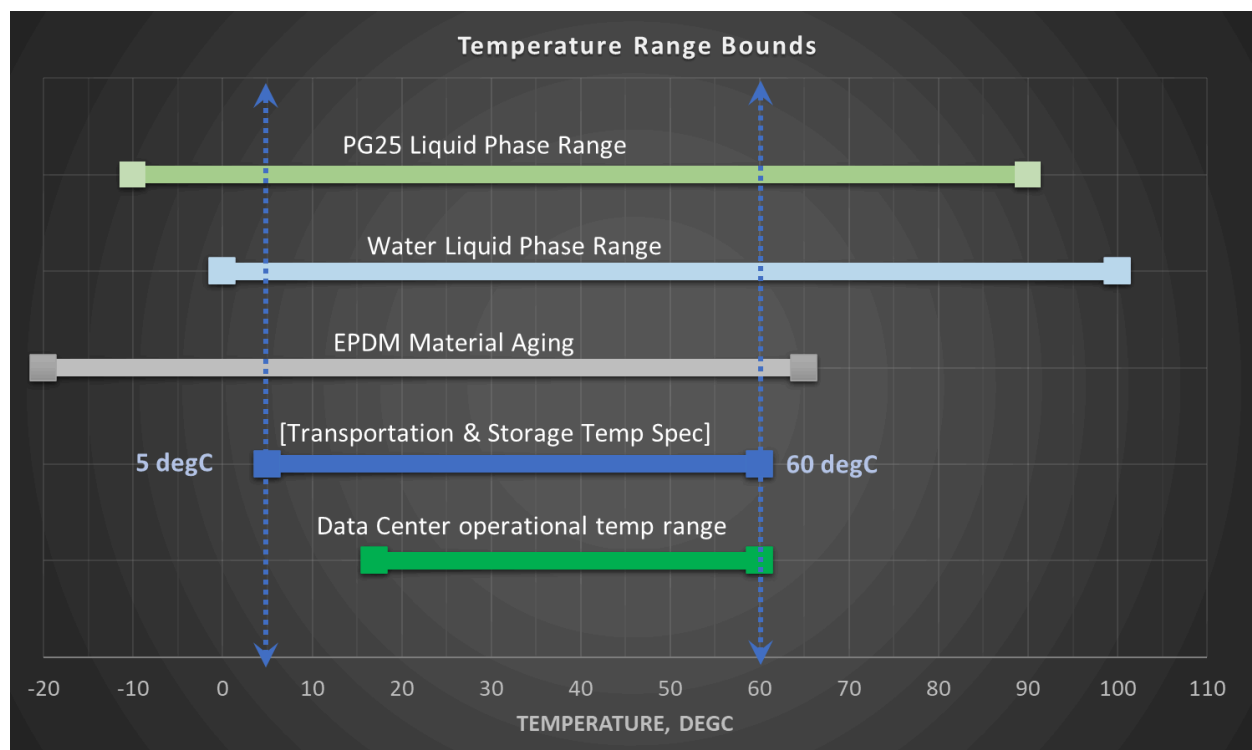
Another potential failure mode is the depletion of corrosion inhibitors during stagnant conditions. Corrosion inhibitors are added to coolants to protect metal components. During initial shipping and storage, a small volume of fluid often remains trapped within the cooling assembly, particularly in copper cold plates. The high copper surface area-to-fluid volume ratio leads to significant initial depletion of inhibitors as they adsorb onto copper surfaces, forming a protective monolayer. This rapid adsorption is crucial for immediate corrosion prevention. Over time, the

depletion rate decreases as the system approaches equilibrium, but ongoing monitoring of inhibitor concentration is necessary to maintain adequate protection. Replenishment may be required to sustain the inhibitor's protective effect throughout the cooling system's lifespan.

With proper passivation of wetted surfaces, use of fresh PG25 coolant with adequate inhibitor levels, and maintenance of appropriate storage conditions (temperature and humidity), the risk of corrosion events due to inhibitor depletion is expected to be low, although the depletion mechanism remains as described above.

## 5. Operational Solutions

### 5.1. Temperature Control for Shipment and Storage



**Figure 2: Recommended temperature control range for shipment and storage**

Considering the coolant liquid phase range, EPDM material aging, and typical data center operating conditions, a non-operational temperature range of 5-60°C is recommended for the

transportation and storage of 'Wet' and 'Partial Wet' materials. This recommendation applies under two scenarios:

1. The liquid-cooled hardware is designed to withstand liquid volume expansion within the 5-60°C range (or potentially a wider range).
2. Transportation and storage conditions are controlled within this range, with a narrower window of 20-25°C recommended for long-term storage.

DOE results indicate that, with appropriate packaging, temperature changes within the hardware occur much more slowly than in the external environment. This thermal inertia allows for environmental temperature excursions outside the recommended range for up to 4-8 hours before significant coolant expansion occurs, reducing the risk of hardware damage.

This temperature range represents a practical balance between operational requirements and resource constraints. Adhering to this guideline allows for the shipment of 'wet' or 'partial wet' products without temperature-controlled transport in many regions, provided weather conditions are favorable.

It should be noted that prolonged storage at elevated temperatures increases corrosion risk due to the formation of thermo-oxidative breakdown by-products. While the 5-60°C range sets the upper and lower bounds, it is generally recommended to maintain long-term storage temperatures at or below 20-25°C.

## 5.2. Jumper hose

Jumper hoses provide an effective solution for managing thermal variations during the shipment of liquid-filled racks. Typically constructed from flexible materials such as EPDM, these hoses create an expansion reservoir to accommodate coolant volume changes due to freezing and subsequent expansion. Jumper hoses can be easily attached or detached, enabling rapid transition between 'partial wet' and 'wet' conditions for shipment or deployment.

For example, consider a liquid-cooled rack containing 20 L of coolant. An empty jumper hose offering an additional 4 L of expansion space can be installed. The jumper hose is initially filled

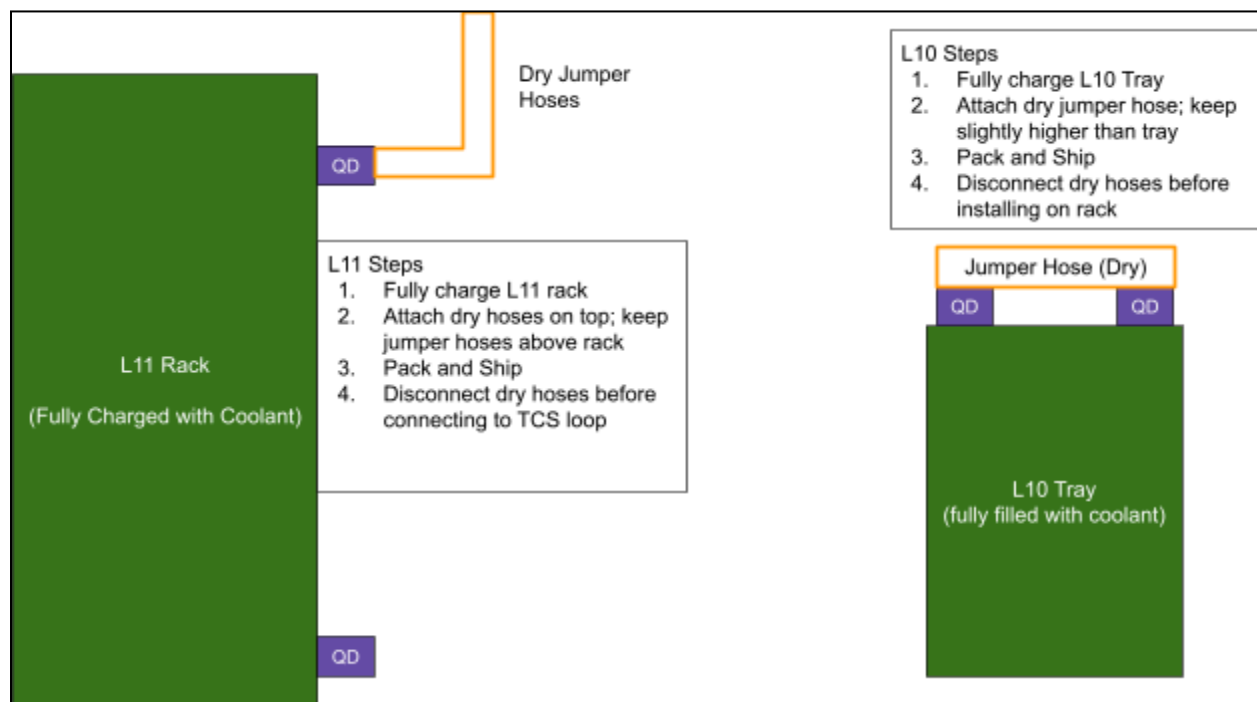
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with air at 1 atm (14.7 psi), which acts as a compressible cushion to absorb coolant volume changes resulting from thermal expansion or contraction. If the coolant freezes and expands by 10%, increasing in volume from 20 L to 22 L, the displaced air in the jumper hose is compressed, reducing the air volume from 4 L to 2 L. According to Boyle's Law, the final air pressure in the jumper hose would rise to 29.4 psi, which remains well within typical system pressure limits (e.g., 75 psi).

See schematic below.



**Figure 3: Illustration of a jumper hose utilized to maintain a closed-loop coolant circuit during rack shipment**

### 5.3. Nitrogen Protection

Nitrogen fill is a standard method for mitigating corrosion risk on wetted metallic surfaces within liquid-cooled hardware. It also serves as a convenient metric for monitoring and determining shelf life.

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A common practice is to charge the liquid cooling assembly with nitrogen to a pressure of 25–50 psi, using vacuum-sealed packaging when applicable, for shipment or long-term storage. Over time, the nitrogen pressure will gradually decay, eventually falling below detectable levels (typically <1 psi). Corrosion processes generally require additional time to initiate after nitrogen is depleted and oxygen becomes present at the wetted surfaces. Thus, residual nitrogen pressure is widely used as a practical indicator of package integrity and remaining shelf life for liquid-cooled hardware.

For properly packaged component or tray level materials, nitrogen protection can provide a shelf life ranging from 3 to 12 months, depending on factors such as joint, seal, and hose material quality, as well as initial charging pressure and storage temperature. Rack-level assemblies, which contain more joints and are often not vacuum-packaged, typically have shorter shelf lives. Additionally, fluctuations in storage temperature and changes in altitude can accelerate nitrogen pressure decay. It is recommended to experimentally determine the shelf life for each specific product configuration.

#### **5.4. How to Drain Effectively**

Compressed clean dry air is commonly used to purge residual liquid from liquid cooling (LC) assemblies. The standard practice involves purging at air pressures below 75 psi for several minutes, typically reducing the residual fluid level to below 5%. It is essential to release the air pressure before transportation, storage, or subsequent processing steps to prevent expansion-related damage due to temperature fluctuations.

## **6. At Scale Considerations**

The decision to ship liquid-cooled hardware in a wet or dry state requires careful evaluation of both technical risks and operational burdens; it is not always a straightforward binary choice. When deploying liquid-cooled hardware at scale, the following factors should be considered:

1. Hardware design robustness
2. Availability of transportation resources

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3. Operational team capacity
4. Speed of delivery (aligned with business objectives)
5. Logistic workflow design

The optimal approach may differ significantly depending on the hyperscale user, deployment site, program requirements, or even among component suppliers within the same program.

For example, in a hypothetical program, the following methods may be adopted:

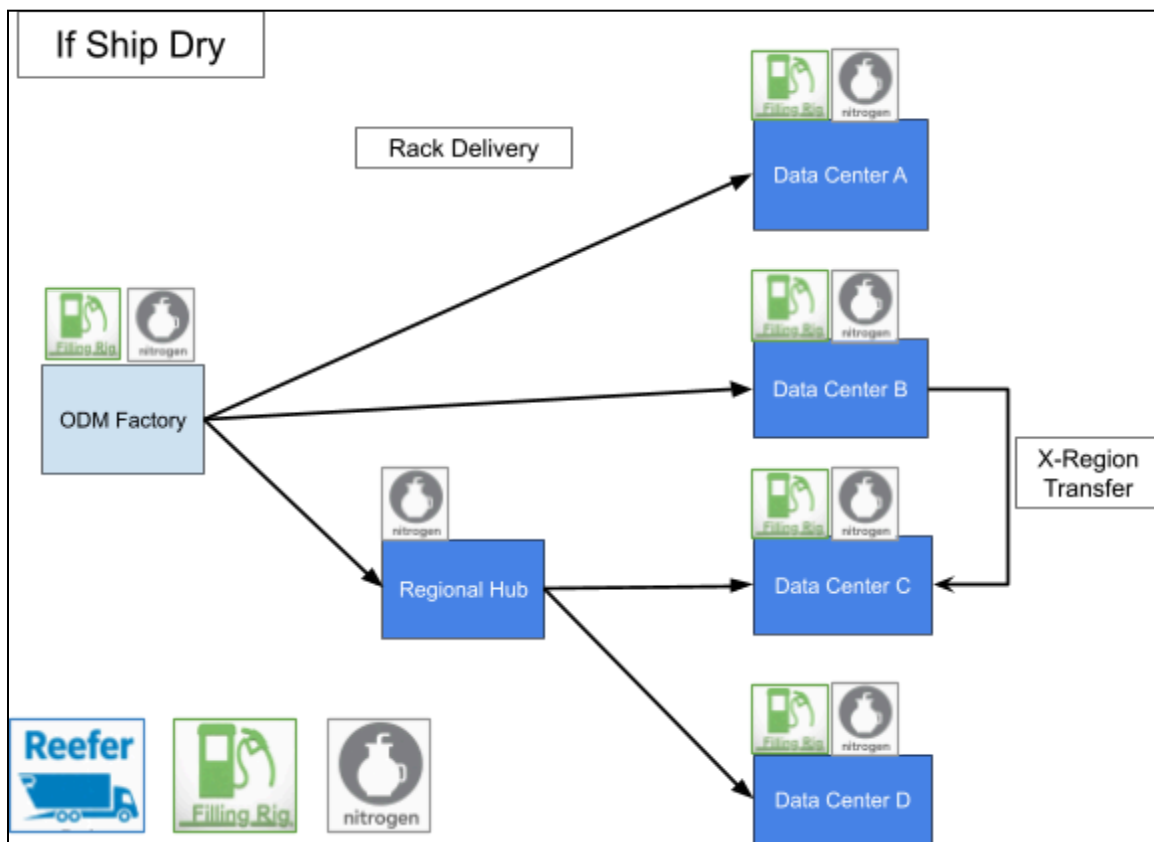
**Table 3 - example of shipping and storing various hardware with different approaches**

|                       | InBound Ship | Storage | OutBound Ship | Outbound due to leakage |
|-----------------------|--------------|---------|---------------|-------------------------|
| L11 Rack              | Wet          | NA      | Wet           | Semi-Dry                |
| Type1 Tray FRU/RMA    | Wet          | Wet     | Wet           | Semi-Dry                |
| Type2 Tray FRU/RMA    | Dry          | Dry     | Semi Dry      | Semi-Dry                |
| Sub component FRU/RMA | Dry          | Dry     | Wet           | Semi-Dry                |

Such decisions was based on unique reasons applied to those elements:

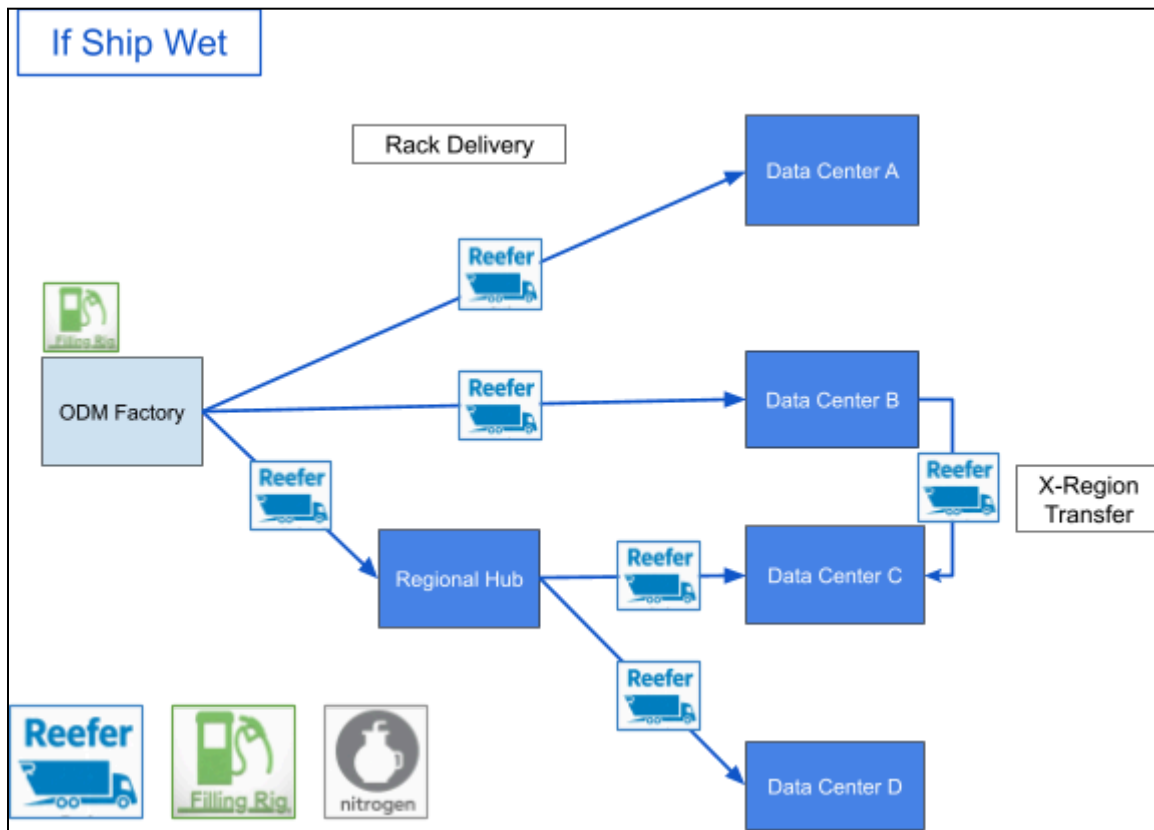
- L11 rack need to land for production quickly
- Type1 Tray has enough flex structure to absorb volume expansion
- Type2 Tray consists of mostly rigid piping and cannot survive volume expansion
- Subcomponent is an off-the-shelf material from a supplier who adopts ship dry globally
- hardware showing leakage cannot be retained wet or dry properly, need to drain liquid and send outbound for repair asap

For data center operators with many destination data centers, the logistics topology can also become a flexible operational model. For example, in a hypothetical scenario, the materials manager can adopt different shipment strategies based on resource availability and any potential bottlenecks.



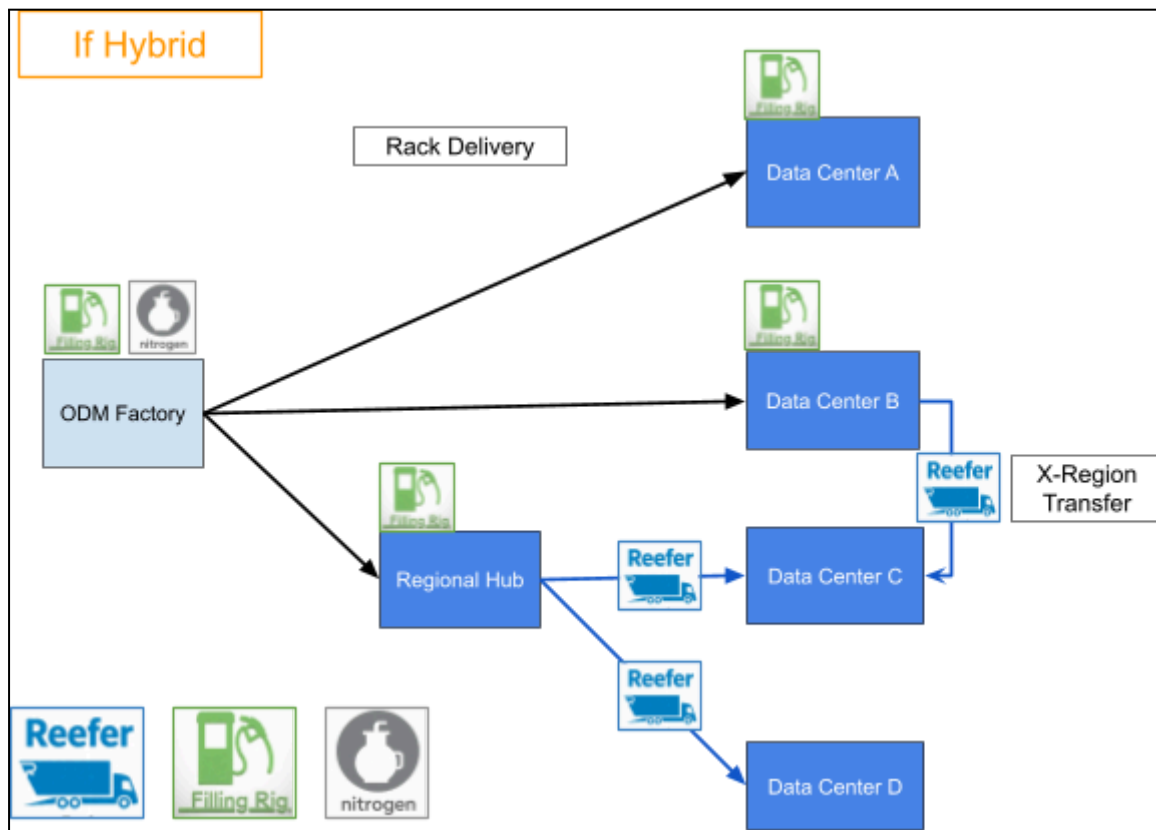
**Figure 4: Example of a dry shipment configuration**

If shipping dry across all regions, it is necessary to establish filling rigs and nitrogen charging equipment at each datacenter. While this approach eliminates the risk of expansion-related damage, it may reduce deployment speed due to the additional steps required for fluid filling and system preparation both before and after transportation.



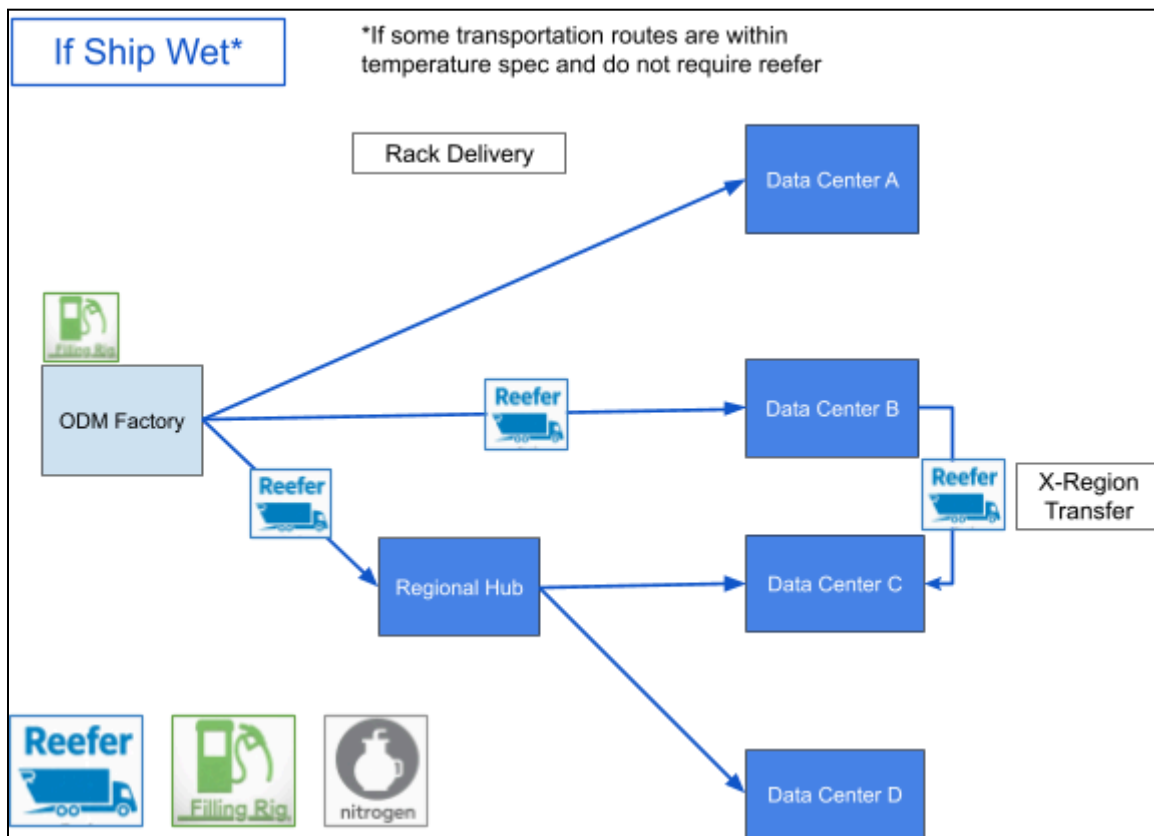
**Figure 5: Example of a wet shipment configuration**

If shipping wet across all regions, reefer (temperature-controlled) transportation must be arranged for all routes. The primary concern with this approach is the availability and reliability of reefer trucks.



**Figure 6: Example of a hybrid shipment configuration**

For a hybrid approach, ship hardware dry from the factory by default. Deploy filling rigs at datacenters with sufficient operational resources, and establish regional hubs to support datacenters with limited operational capacity.



**Figure 7: Example of wet shipment with or without reefer**

If shipping is wet, instead of utilizing reefers for all shipments, determine the need for temperature-controlled transportation based on specific routes and weather data. This approach reduces resource requirements for the factory, transportation, and post-arrival operations, but necessitates more detailed analysis of transportation scenarios and comprehensive data collection.

While there is a strong desire to standardize shipment protocols across all programs, products, and destinations within the data center operator's supply chain, achieving full homogeneity in the near term remains challenging and is contingent on the maturity of the broader ecosystem. As products from individual components to fully integrated racks are increasingly engineered to withstand both dry and wet shipping, and as operational resources (such as refrigerated transport, on-site fluid filling, and nitrogen purging) become more widely available, the industry will gain greater flexibility.

This evolution will enable a future where:

- Shipment mode (wet or dry) can be selected based on liquid cooling hardware requirements and global logistics strategies.
- Users can choose shipment methods best suited to their operational needs, reducing competition for limited resources.
- The industry achieves more proficient handling of liquid-cooled hardware, ensuring that transportation and storage do not become bottlenecks for product launches.
- By embedding flexibility and resilience into the supply chain, data center operators and their partners will be better positioned to support the rapid growth and diversification of liquid-cooled deployments worldwide.

## 7. Conclusion

Coolant expansion damage and corrosion are the two primary risks associated with non-operational transportation and storage of liquid-cooled hardware.

Based on current observations, our recommendations for different shipping and storage modes are as follows. Please note that variations in product design, coolant type, and user requirements may lead to significant differences in optimal practices.

**Table 4 - comparison between wet and dry approaches**

| Mode                                  | Wet                                              | Dry                                              |
|---------------------------------------|--------------------------------------------------|--------------------------------------------------|
| Definition                            | Filled with coolant                              | Minimum coolant residual                         |
| Risk                                  | Expansion Damage<br>Inhibitor Depletion          | Corrosion                                        |
| Solutions                             | Temperature Control<br>Jumper Hose               | Drain with compressed air<br>Nitrogen Protection |
| Shipment Timewindow<br>Recommendation | Below 1 month                                    | Below 1 month                                    |
| Typical Shelf Life                    | 9~12months                                       | 3~12months                                       |
| Temp Limit<br>Recommendation          | Absolute: 5~60degC<br>Storage Average: 20~25degC | Absolute: None<br>Storage Average: None          |

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Semi-wet and semi-dry modes offer flexibility for expedited material delivery but are not recommended for large-scale deployment or long-term storage.

Multiple strategies exist to address the transportation and storage requirements of liquid-cooled hardware. Wet and dry modes each present distinct resource demands and operational efficiencies; users may select a single mode or combine multiple approaches to facilitate scalable delivery.

## 8. Glossary

DOE - Design of Experiments

EPDM - Ethylene Propylene Diene Monomer

LC - Liquid Cooling

MEP - Mechanical, Electrical and Plumbing

NPI - New Product Introduction

ODM - Original Design Manufacturer

PG25 - Propylene Glycol (25 vol%)

RMA - Return Merchandise Authorization

Reefer - Refrigeration Truck

## 9. References

### 9.1. Web resources

[1] <https://webstore.iec.ch/en/publication/29793>

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## 11. About Open Compute Foundation

The Open Compute Project (OCP) is a collaborative Community of hyperscale data center operators, telecom, colocation providers and enterprise IT users, working with the product and solution vendor ecosystem to develop open innovations deployable from the cloud to the edge. The OCP Foundation is responsible for fostering and serving the OCP Community to meet the market and shape the future, taking hyperscale-led innovations to everyone. Meeting the market is

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accomplished through addressing challenging market obstacles with open specifications, designs and emerging market programs that showcase OCP-recognized IT equipment and data center facility best practices. Shaping the future includes investing in strategic initiatives and programs that prepare the IT ecosystem for major technology changes, such as AI & ML, optics, advanced cooling techniques, composable memory and silicon. OCP Community-developed open innovations strive to benefit all, optimized through the lens of impact, efficiency, scale and sustainability. Learn more at [www.opencompute.org](http://www.opencompute.org).